

1. Administrative & Study Identification

Full Study Title: A Phase III, Multicenter, Randomized, Open-Label Study of Atezolizumab (Anti-PD-L1 Antibody) Plus Bevacizumab Versus Active Surveillance as Adjuvant Therapy in Patients With Hepatocellular Carcinoma at High Risk of Recurrence After Surgical Resection or Ablation **Short Title/Acronym:** IMbrave050 / A Study of Atezolizumab Plus Bevacizumab Versus Active Surveillance as Adjuvant Therapy in Patients With Hepatocellular Carcinoma at High Risk of Recurrence After Surgical Resection or Ablation **Protocol Number:** W041535 **NCT Number:** NCT04102098 **Posting ID:** 514211514805643384 **Trial Status:** Active, not recruiting **Sponsor Name:** Hoffmann-La Roche / Genentech (Company: Roche) **Investigational Product:** Atezolizumab (Tecentriq, Anti-PD-L1 Antibody) plus bevacizumab (Avastin, Vegzelma, Alymsys, Zirabev, Mvasi) **Phase of Development:** Phase III **Medical Monitor:** [Roche/Genentech Clinical Operations Lead]

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of adjuvant therapy with atezolizumab plus bevacizumab compared with active surveillance in improving recurrence-free survival (RFS) in participants with completely resected or ablated hepatocellular carcinoma (HCC) who are at high risk for disease recurrence. **Secondary Objectives:** To evaluate overall survival (OS), recurrence-free survival (RFS) as determined by the investigator, time to recurrence (TTR), time to extrahepatic spread (EHS) or macrovascular invasion, RFS in the PD-L1-high subgroup, serum concentration of atezolizumab, and the incidence of anti-drug antibodies.

2.2 Background & Rationale To address the high risk of recurrence in early-stage Hepatocellular Carcinoma (HCC). Historically, more than 70% of people with early-stage HCC experience disease recurrence after surgery, which is associated with a poor prognosis and shorter survival. Dual PD-L1/VEGF blockade (atezolizumab + bevacizumab) has shown significant benefits in unresectable HCC and is hypothesized to reduce HCC recurrence by establishing a more immune-favorable microenvironment following curative resection or ablation.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Surveillance **Blinding:** Open-label **Randomization:** 1:1 using permuted blocks with a block size of 4

3.2 Planned Enrollment & Duration Target **\$N\$:** 668 patients **Number of Sites:** 134 hospitals and medical centers across 26 countries **Estimated Study Start:** Dec 31, 2019 (First patient randomized) **Estimated Primary Completion:** Oct 21, 2022 (Prespecified interim analysis data cutoff)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participants with a first diagnosis of HCC who have undergone either a curative resection or ablation (radiofrequency ablation [RFA] or microwave ablation [MVA] only) within 4-12 weeks prior to randomization. 2. Documented diagnosis of HCC that has been completely resected or ablated (RFA or MVA only) and high risk for HCC recurrence after resection or ablation. 3. Absence of major macrovascular invasion (except Vp1/Vp2) and absence of extrahepatic spread as confirmed by CT or MRI scan of the chest, abdomen, pelvis, and head prior to and following curative procedure. 4. Full recovery from surgical resection or ablation within 4 weeks prior to randomization. 5. For patients who received post-operative transarterial chemoembolization (TACE): full recovery from the procedure within 4 weeks prior to randomization. 6. For patients with resected HCC, availability of a representative baseline tumor tissue sample. 7. ECOG Performance Status of 0 or 1, Child-Pugh Class A status, and adequate hematologic and end-organ function. 8. For women of childbearing potential: agreement to remain abstinent (refrain from heterosexual intercourse) or use contraceptive methods. For men: agreement to remain abstinent or use a condom, and agreement to refrain from donating sperm.

4.2 Exclusion Criteria 1. Pregnant or breastfeeding, or intending to become pregnant during the study or within 5 months after the final dose of atezolizumab or within 6 months after the final dose of bevacizumab. Women of childbearing potential must have a negative serum pregnancy test result within 14 days prior to Day 1 of Cycle 1. 2. Known fibrolamellar HCC, sarcomatoid HCC, or mixed cholangiocarcinoma and HCC. 3. Evidence of residual, recurrent, or metastatic disease at randomization. 4. Clinically significant ascites or history of hepatic encephalopathy. 5. Prior bleeding event due to untreated or incompletely treated esophageal and/or gastric varices within 6 months prior to randomization. 6. Have received more than 1 cycle of adjuvant TACE following surgical resection. 7. Active or history of autoimmune disease or immune deficiency. 8. History of idiopathic pulmonary fibrosis, organizing pneumonia, drug-induced pneumonitis, or idiopathic pneumonitis, or evidence of active

pneumonitis on screening chest CT scan. 9. Significant cardiovascular disease within 3 months prior to Day 1 of Cycle 1, unstable arrhythmia, or unstable angina. 10. History of malignancy other than HCC within 5 years prior to screening, with the exception of malignancies with a negligible risk of metastasis or death. 11. Active tuberculosis. 12. Co-infection with HBV and HCV, or co-infection with HBV and hepatitis D viral infection. 13. Clinical significant uncontrolled or symptomatic hypercalcemia. 14. Any treatment for HCC prior to resection or ablation, including systemic therapy and locoregional therapy such as TACE. 15. Treatment with systemic immunostimulatory or immunosuppressive agents. 16. Inadequately controlled arterial hypertension, history of hypertensive crisis or hypertensive encephalopathy, or significant vascular disease. 17. Evidence of bleeding diathesis or significant coagulopathy, current or recent use of aspirin or full-dose oral or parenteral anticoagulants. 18. Core biopsy within 3 days of Day 1 of Cycle 1. 19. History of GI fistula, GI perforation, or intra-abdominal abscess. 20. Serious non-healing or dehiscing wound, or major surgical procedure within four weeks. 21. Chronic daily treatment with a non-steroidal anti-inflammatory drug. 22. Any other disease, metabolic dysfunction, physical examination finding, or clinical laboratory finding that contraindicates the use of an investigational drug, may affect the interpretation of the results, or may render the patient at high risk from treatment complications.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Atezolizumab 1200 mg plus bevacizumab 15 mg/kg every 3 weeks **Route of Administration:** Intravenous (IV) **Duration of Treatment:** 17 cycles (Maximum of 12 months)

5.2 Concomitant & Prohibited Therapy Permitted: Standard-of-care supportive medications as dictated by site protocols. **Prohibited:** Systemic immunostimulatory or immunosuppressive agents; current or recent use of aspirin or full-dose oral/parenteral anticoagulants; chronic daily treatment with NSAIDs.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, CT/MRI confirmation of complete resection/ablation, Negative pregnancy test within 14 days (if applicable)
Baseline	Day 0	Randomization, First Dose (for experimental arm), Safety Labs	Interim Every 3 Weeks Treatment administration (Q3W), Safety and Toxicity Tracking, Efficacy Assessment
End of Study	12 Months	Final	

treatment cycle, transition to extended Survival Follow-up for RFS and OS |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Recurrence-Free Survival (RFS), as Determined by IRF, defined as the time from randomization to the first documented occurrence of intrahepatic or extrahepatic HCC, or death from any cause (whichever occurs first).

Measurement Method: Independent Review Facility (IRF) assessment.

7.2 Secondary & Exploratory Endpoints * Overall Survival (OS). * RFS as Determined by the Investigator. * Time to Recurrence (TTR). * Time to extrahepatic spread (EHS) or macrovascular invasion. * Recurrence-free survival (RFS) specifically among patients in the PD-L1-high subgroup.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of treatment-related AEs, specifically tracking Grade 3 or 4 adverse events, bleeding events, and thromboembolic events. **Serious**

Adverse Events (SAEs): Evaluated according to standard 24-hour reporting timelines in the safety-evaluable population. **Data**

Monitoring Committee (DMC): External independent Data Monitoring Committee oversight for standard oncology toxicity reporting.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intention-to-Treat (ITT) population for primary efficacy endpoints. Safety-evaluable population for toxicity assessment. **Sample Size Justification:** $N=668$, powered to detect a statistically significant shift in recurrence-free survival (RFS) comparing the combination therapy vs. active surveillance using stratified Cox proportional-hazards models.

Handling of Missing Data: Standard survival analysis algorithms utilizing Kaplan-Meier curves and censoring for patients lost to follow-up or without documented recurrence by data cutoff.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Scheduled onsite and remote monitoring for Phase III clinical trial protocol adherence, managed by Hoffmann-La Roche.

Audit Trail: Standardized eCRF locking and blinded Independent Review Facility (IRF) imaging audits.

11. Study Identifiers & Governance

Primary ID: W041535 **Registry Number:** NCT04102098 **Sponsor/Lead Organization:** Hoffmann-La Roche / Genentech **Study Title:** IMbrave050 **Official Regulatory Title:** A Phase III, Multicenter, Randomized, Open-Label Study of Atezolizumab (Anti-PD-L1 Antibody) Plus Bevacizumab Versus Active Surveillance as Adjuvant Therapy in Patients With Hepatocellular Carcinoma at High Risk of Recurrence After Surgical Resection or Ablation

12. Clinical Framework (HCC Specific)

Primary Condition: Hepatocellular Carcinoma (HCC) (Disease Area: Carcinoma, Hepatocellular) **Diagnosis Threshold:** Complete surgical resection or ablation with a high risk of disease recurrence **Medical Methodology:** Adjuvant Dual PD-L1/VEGF Blockade Immunotherapy **Optimization Target:** Improvement of recurrence-free survival (RFS) and overall survival (OS)

13. Study Procedures & Methodology

Management Pathway: 12-month Adjuvant Atezolizumab + Bevacizumab vs. Active Surveillance **Education Protocol (HCP):** Site initiation training covering IV infusion protocols and stringent AE (e.g., bleeding, hypertension) monitoring **Education Protocol (Patient):** Informed consent, post-surgical recovery timelines, and active surveillance schedule alignment **Quality Audit Mechanism:** Central blinded Independent Review Facility (IRF) imaging tracking for recurrence

14. Observation & Follow-Up Schedule

Total Duration: 12 months (17 cycles) active intervention, followed by long-term survival monitoring **Onsite Visit Cadence:** Every 3 weeks (Q3W) matching the IV infusion and monitoring cycle **Remote Monitoring Frequency:** Intermittent remote checks during the long-term active surveillance period. **Checklist Review Interval:** End of every 3-week cycle.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[07-Apr-2026]				
Clinical Operations Lead	[Name]	_____	[07-Apr-2026]				Lead
Statistician	[Name]	_____	[07-Apr-2026]				